

Saptarshi Maji, PhD

IPMC-CNRS, Valbonne, France, 06560

+ (33) 074 555 8977

maji@ipmc.cnrs.fr

saps018.github.io

Post-doctoral fellow

Education

- 2018 – 2024 Ph. D. scholar in Department of Biological Sciences (DBS), *IISER Kolkata*.
2013 – 2018 BS-MS dual degree in Biological Sciences, *DBS, IISER Kolkata*, CGPA **8.09/10**.
2013 Higher Secondary Examination, XIIth standard, *WBCHSE Board*, Scored **89%** in science.
2011 Secondary Examination, Xth standard, *WBBSE Board*, Scored **95%** in science.

Research Experiences

Post-doctoral experience

- 2024 – Curr Understanding the role of Zinc in regulating the function of a novel keratinocyte differentiation marker, under supervision of *Prof. Bruno Antonny*, IPMC-CNRS, France.

Doctoral thesis

- 2018 – 2024 Functional and Temporal characterisation of Wilson Disease Protein ATP7B at the Lyso-somes, under supervision of *Prof. Arnab Gupta*, DBS, *IISER Kolkata*.

Master's thesis

- 2017 – 2018 Regulatory mechanism of copper transporters hCTR1 and ATP7B for maintaining copper homeostasis, under supervision of *Prof. Arnab Gupta*, DBS, *IISER Kolkata*.

Summer Internships

- Summer Over Expression and Purification of Recombinant PCBP2 Protein to Generate Antibody
2016 and Study Its Interaction with Coxsackie virus B3 RNA, under supervision of *Prof. Saumitra Das*, Molecular and Cellular Biology, Indian Institute of Science (IISc) Bangalore.
Summer Survey of Endosymbiont Infection among Indian Arthropods, under supervision of *Dr. Rhi-*
2015 *toban RayChoudhury*, Department of Biological Sciences, *IISER Mohali*.
Summer Preventing Racemization of Chiral Schiff-base Ligands and Cu(II) Complexes in Solution
2014 to synthesize new architecture for chiral recognition, under supervision of *Prof. Manabendra Ray*, Department of Chemistry, Indian Institute of Technology (IIT) Guwahati.

Skills

Experimental Experienced with animal cell culture (2D and 3D), genetic engineering and CRISPR technology, general biochemistry and molecular biology techniques, super-resolution microscopy, multi-omics analysis, protein condensate, and lipid biochemistry.

Software Proficient in **ImageJ**, **R**, **PYTHON**, **MaxQuant**, **CellProfiler**, **HTML**, and **LaTeX**

Publications

- Brindaban Maji†, **Saptarshi Maji*†**, Arpan Narayan Roy, Ruchi Singh, Vikas Yadav, Borkar Aditya Ashok, Bratati Dutta, Sandip Sarkar, Avijit Majumder, Mainak Chakrabarty, Swastika Dey, Tisha Dash, Pruthwiraj, Aidan T. Pezacki, Christopher J. Chang, Amitava Das, Soumik Siddhanta, Arnab Gupta*; **Copper imbalance in Wilson disease drives dysregulation of the lysosome-mTOR-cholesterol axis in hepatocytes**, (*communicated*, *METABOLISM-D-26-01535*) (*Corresponding author)
- Sharon Mary Jose†, **Saptarshi Maji†**, Mrityika Paul, Chirag Singhal, Sreeramaiah N Gangappa, Aayatti Mallick Gupta, Arnab Gupta; **Comparative analysis of key Copper-ATPase domains sheds light on evolutionary divergence in human homologues ATP7A and ATP7B**, *J. Cell Sci.* (2025)
- Saptarshi Maji**, Marinella Pirozzi, Raturaj, Raviranjana Pandey, Tamal Ghosh, Santanu Das, Arnab Gupta; **Copper-independent lysosomal localization of the Wilson disease protein ATP7B**, *Traffic* (2023)

4. **Saptarshi Maji**, Ruturaj, Raviranjana Pandey, Santanu Das, Arnab Gupta; **Trans-Golgi Network-Lysosome proximity sites govern trafficking of Wilson disease protein ATP7B in hepatocytes**, *CellBio23 (P2079)*
5. Basudeb Mondal†, Abinash Padhy†, **Saptarshi Maji†**, Arnab Gupta, Sayam Sen Gupta; **Dual stimuli-responsive cross-linked nanoassemblies from an amphiphilic mannose-6-phosphate based tri-block copolymer for lysosomal membrane permeabilization**, *Biomater. Sci. (2023)*
6. Sumanta Kar†, Samarpita Sen†, **Saptarshi Maji**, Deepashri Saraf, Ruturaj, Rupam Paul, Sohini Dutt, Basudeb Mondal, Enrique Rodriguez-Boulan, Ryan Schreiner, Durba Sengupta, Arnab Gupta; **Copper(II) import and reduction are dependent on His-Met clusters in the extracellular amino terminus of human copper transporter-1**, *J. Biol. Chem. (2022)*
7. Santanu Das, **Saptarshi Maji†**, Ruturaj†, Indira Bhattacharya, Tanusree Saha, Nabanita Naskar, Arnab Gupta; **Retromer retrieves the Wilson disease protein ATP7B from endolysosomes in a copper-dependent manner**, *J. Cell Sci. (2020)*
8. Taniya Mandal†, Sumanta Kar†, **Saptarshi Maji†**, Samarpita Sen†, Arnab Gupta; **Structural and Functional Diversity Among the Members of CTR, the Membrane Copper Transporter Family**, *J.Membr.Biol.(2020)*
9. Subhendu Kumar Chatterjee†, Sumanta Kar†, Siddhanta Nikte†, Tisha Dash†, **Saptarshi Maji**, Tanmoy Ghosh, Durba Sengupta, Bidisha Sinha, Arnab Gupta; **Non-canonical regulation of the plasma membrane copper transporter CTR1 through modulation of membrane mechanical properties**, *MBoC (2026)*
10. Raviranjana Pandey, Arpan Narayan Roy, Sandip Sarkar, Rakiba Rohman, Kaustav Chakraborty, Rupa Bar-gakshatriya, Sanjana Pandey, Pruthwiraj, Debosmita Bhattacharya, Saurav Kumar, **Saptarshi Maji**, Aidan T Pezacki, Sumit K Pramanik, Christopher J Chang, Ashima Bhattacharjee, Neelanjana Sengupta, Amitava Das, Arnab Gupta; **Repurposing Melatonin in dual-mode for Wilson disease therapy as a Copper Chelator and an antioxidant agent**, *Redox Biol. (2025)*
11. Rupam Paul, Adrija Chakraborty, Suman Samanta, Swastika Dey, Raviranjana Pandey, **Saptarshi Maji**, Aidan T. Pezacki, Christopher J. Chang, Rupak Datta, Arnab Gupta; **Leishmania major-induced alteration of host cellular and systemic copper homeostasis drives the fate of infection**, *Commun. Biol. (2024)*
12. Ruturaj, Monalisa Mishra, Soumyendu Saha, **Saptarshi Maji**, Enrique Rodriguez-Boulan, Ryan Schreiner, Arnab Gupta; **Regulation of apico-basolateral trafficking polarity of homologous Copper-ATPases ATP7A and ATP7B**, *J. Cell Sci. (2023)*
13. Kaustav Chakraborty, Santanu Das, Anusree Pal, **Saptarshi Maji**, Bhawana Rai, Arnab Gupta, Ashima Bhattacharjee; **Wilson disease-causing mutations in the carboxyl terminus of ATP7B regulates its localization and Golgi exit selectively in the unpolarized cells**, *Metallomics (2023)*
14. Santanu Das, Ameena Mohammed, Taniya Mandal, **Saptarshi Maji**, Jay Verma, Ruturaj, Arnab Gupta; **Polarized trafficking and copper transport activity of ATP7B: A mutational approach to establish genotype–phenotype correlation in Wilson disease**, *Hum. Mutat. (2022)*
15. Rupam Paul, Sourav Banerjee, Samarpita Sen, Pratiksha Dubey, **Saptarshi Maji**, Anand K Bachhawat, Rupak Datta, Arnab Gupta; **A novel leishmanial copper P-type ATPase plays a vital role in parasite infection and intracellular survival**, *J. Biol. Chem. (2021)*
16. Moumita Maji, Sourav Acharya, **Saptarshi Maji**, Kallol Purkait, Arnab Gupta, Arindam Mukherjee; **Differences in Stability, Cytotoxicity, and Mechanism of Action of Ru(II) and Pt(II) Complexes of a Bidentate N₂O Donor Ligand**, *Inorg. Chem. (2020)*

†equal contribution

Conference Preceedings

- 2024 Invited as **guest speaker** in **Amity Institute of Biotechnology, Amity University, Kolkata, India**
- 2023 Selected for **poster presentation** in **Cell Bio 23** organised by **ASCB|EMBO, Boston, USA**
Selected for **poster presentation** in **OBMT 2023 conferenc, NBRC, India**
Selected for **oral presentation** in **Organelar Channels and Transporters Gordon Research Conference 2023, Barcelona, Spain**
Selected for **poster presentation** in **Frontiers in Modern Biology (FiMB), IISER Kolkata**
- 2022 Selected for **poster presentation** in **91st annual meeting of Society of Biological Chemists**
Selected for **oral presentation** in **Society of Biological Chemists (I) Kolkata chapter**
- 2021 Attended **Molecular Membrane Biology meeting, Online**

- 2020 Selected for **International Symposium on Cell Surface Macromolecules**, IISER Pune
- 2019 Selected for **Multiscale Simulation & Mathematical Modelling of Complex Biological Systems** conference and pre-conference workshop, JNU
- 2018 Selected for **poster presentation** in **Frontiers in Modern Biology (FiMB)**, IISER Kolkata
- 2017 Attended **Advances in Biological Sciences**, IISER Kolkata
- 2015 Attended **Frontiers in Modern Biology (FiMB)**, IISER Kolkata
- 2013 Attended **SPIE winter school on bio-photonics**, IISER Kolkata
- 2011 Selected in **DST-INSPIRE science camp internship** during high school

Fellowships and Awards

- 2026 Seal of Excellence, from **Marie Skłodowska-Curie Actions**, European Commission
- 2023 **International Travel Scholarship** by **SCIENCE & ENGINEERING RESEARCH BOARD (SERB)**
International Travel Scholarship by **Research and Development**, IISER Kolkata
Camillo Golgi best poster award in the **Organelle Biogenesis & Membrane Trafficking** conference (OBMT) 2023, NBRC, India
Gold Medal for IISER Kolkata Team in the **iGEM Grand Jamboree 2023**, Paris expo, France
- 2020-2023 **Joint CSIR-UGC NET SRF scholarship** by Govt. of India
- 2018-2020 **Joint CSIR-UGC NET JRF scholarship** by Govt. of India
- 2013-2018 **INSPIRE Fellowship** by **Department of Science and Technology**, Govt. of India

Teaching Experiences

- 2018-2024 Mentored several summer interns, undergraduate and fresh graduate students during PhD tenure
- 2023 **Student mentor** of **iGEM 2023 IISER Kolkata Undergrad Team**
- June, '22 **Module Instructor** of **Microscopy and Image analysis** for **Karyashala (Workshop)** on advanced bio-analytical methods and applications (BAW-2022)
- Autumn, '22 **Teaching Assistant** for 3rd year BSMS Students for the **Cell Biology and Imaging Laboratory**
- Autumn, '19 **Teaching Assistant** for 4th year BSMS Students for the **Advanced Physiology** course
- Autumn, '17 **Teaching Assistant** for 1st year BSMS Students for the **Basic Life Sciences** course

Professional activities

1. Reviewed for **Apoptosis**, Springer Nature
2. Reviewed for **Cell Biology and Toxicology**, Springer Nature
3. Reviewed for **Bio-protocol Journal**
4. Member of **American Society for Cell Biology (ASCB)**
5. Official Nominator for **VinFuture Prize 2026**, Vietnam

Extracurricular Activities

1. Member of **International Relation and Outreach (IRO)**, IISER Kolkata, India.
2. **Guest Editor** and **content writer** of '**Cogito137**', science communication magazine of IISER Kolkata, India.
3. **Founder member** of '**Art for a Cause**', an NGO that provided aid to Covid affected people
4. **Organiser** and **Volunteer** of several conferences in **IISER Kolkata**
5. **Event photographer** for FIMB 2018, IISM (Inter IISER Sport Meet) 2017, and IICM (Inter IISER Cultural Meet) 2018.
6. Write poems and short stories in Bengali. Acted as **editor** in several literary magazines.
7. Also interested in painting, photography, traveling, and trekking

For Reference

1. Prof. Arnab Gupta, DBS, IISER Kolkata, India, email: arnab.gupta@iiserkol.ac.in
2. Dr. Bruno Antonny, IPMC-CNRS, France, email: antonny@ipmc.cnrs.fr
3. Prof. Bidisha Sinha, DBS, IISER Kolkata, India. email: bidisha.sinha@iiserkol.ac.in
4. Prof. Rupak Datta, DBS, IISER Kolkata, India. email: rupakdatta@iiserkol.ac.in